CONCEPT NOTE: Digital Water Footprint

1) Project Name

Digital Water Footprint: Quantifying the Invisible Water Consumption of Social Media Usage

2) Overview

This project makes visible the hidden environmental cost of everyday social media usage by quantifying its water consumption. Every time a user watches a video, scrolls through a feed, or sends a message, data is processed and transmitted through large-scale data centres. These facilities require significant amounts of water for cooling systems, yet this cost remains entirely invisible to the end user.

The project calculates each user's "digital water footprint" by combining three data layers: user screen time per application, data traffic generated per platform (GB/hour), and the Water Usage Effectiveness (WUE) coefficient of the data centre infrastructure used by each platform. The formula applied is: Water (L) = Screen Time (hr) × GB/hr × Litre/GB.

Exploratory Data Analysis (EDA) was conducted on a merged dataset of 2,434 records across 10 social media platforms. Key findings revealed that the average digital water footprint per usage record is 0.1186 litres, with a total footprint of 288.72 litres across all records. Instagram and TikTok emerged as the highest-consuming platforms, averaging 0.3745 L and 0.2600 L per session respectively.

This project is aligned with SDG 6 (Clean Water and Sanitation), specifically Target 6.4, which calls for increasing water-use efficiency across all sectors. By revealing the water cost embedded in digital behaviour, the project empowers users to make more informed and sustainable choices. SDG 6 (Clean Water and Sanitation): aimed at increasing water-use efficiency.

3) Background

While digital platforms are often marketed as "paperless" and eco-friendly, they hide a significant physical resource demand. Data centers are among the most energy-intensive and water-dependent infrastructures globally. As the consumption of high-definition video and social media content increases, the amount of water required to cool these centers reaches critical levels.

Most users are unaware that their digital activities deplete physical resources. This study bridges this awareness gap by modeling the relationship between user behavior (Screen Time) and physical resource depletion (Liters of Water). By revealing the water cost difference between high-definition streaming and simple messaging, the research explains the environmental impact of digital consumption through concrete data.

4) Key Objectives / Business Objectives

a) Research Questions:

1. How much water is consumed per user per session as a result of social media activity, and how does this vary by platform?

2. Which platforms contribute most to the digital water footprint, and what infrastructure factors (WUE, data rate) drive this difference?

3. What is the correlation between screen time duration and water consumption, and how strong is this relationship across different platforms?

b) Key Steps:

1. Collect and merge three datasets: user screen time records, per-application data traffic rates (GB/hour), and data centre WUE coefficients by platform.

2. Calculate the digital water footprint for each user-platform-session using the formula: Water (L) = Screen Time (hr) × GB/hr × Litre/GB.

3. Conduct Exploratory Data Analysis (EDA) to identify trends, platform-level differences, missing value patterns, and correlations.

45) Methods and Workflow

The study follows a structured data processing pipeline implemented in Python:

a) Data Collection:

- **Screen Time:** Raw logs of daily application usage duration (minutes).
- **Data Intensity Coefficients:** Hourly GB consumption rates for each specific platform (e.g., YouTube, Instagram).
- **Water Usage Coefficients:** The amount of water consumed per 1 GB of data processed, derived from official environmental reports (WUE data from Google and Meta).

b) Data Preprocessing & Feature Engineering:

1. **Normalization:** Standardizing application names and cleaning usage logs for consistency.
2. **Traffic Calculation:** Engineering the Total_GB_Usage feature using the formula: (screen_time_min * app_gb_coefficient) / 60.
3. **Final Quantification:** Calculating Total_Water_Footprint by multiplying the data volume with platform-specific water usage factors.

c) Modelling & Analysis:

1. **Correlation Analysis:** Utilizing a Pearson Correlation Matrix to examine the statistical relationship between Screen Time, Data Volume, and Water Consumption.
2. **Intensity Benchmarking:** Calculating the "Litre per Minute" metric for each platform to evaluate and compare resource efficiency.

d) Deliverables

1. **Final Analyzed Dataset:** A comprehensive CSV file containing water footprint calculations for 10 different platforms.
2. **EDA Visualizations: * Bar Chart:** A comparative analysis of water footprint per minute, identifying YouTube (0.0033 L/min) as the highest impact platform.
    - **Heatmap:** A correlation matrix proving a **0.78 relationship** between data volume and water consumption.
3. **Statistical Insights:** Scientific evidence proving that data density (resolution) is a more critical driver of water scarcity than total usage time.

Begüm Erdoğan

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